CENTER FOR ECONOMIC DEVELOPMENT RESEARCH · RESEARCH MEMORANDUM

Community Asset Quality and Constraint Bundling in Rural South Georgia

An empirical test of David Bridges' systems-integration hypothesis across the 99 counties of Georgia Regional Commissions 6–12

Prepared for	Georgia Regional Commissions 6–12
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Study area	99 counties — Middle Georgia, CSRA, River Valley, Heart of Georgia–Altamaha, Southwest Georgia, Southern Georgia, Coastal
Evidence base	92 indicators from 10 federal data programs, mapped to 29 community-asset sub-assets in 8 domains
Companion	Interactive dashboard: county scorecards, constraint matrix, bundle profiles, sequencing analysis

Executive summary

The question. Communities across Regional Commissions 6–12 describe a common list of constraints — workforce, housing, healthcare, infrastructure, local capacity. The list is familiar from the Aspen Institute, UN and other literatures, and practitioners report that the constraints compound. David Bridges' systems-integration hypothesis goes further: because the constraints compound, they cannot be relieved one at a time — they have to be addressed together, and a programme that sequences them will underdeliver. This memo tests that hypothesis against public data: are the reported constraints real, do they bundle, and if they bundle, can they be sequenced?

What we built. An Asset Quality Score for each of the 99 counties, assembled from 92 indicators drawn from the American Community Survey, County Business Patterns, the 2022 Census of Governments, LEHD commuting flows, BLS labour statistics, Census population estimates and building permits, CDC PLACES, CMS hospital records and TIGER road geography. Each indicator is assigned to exactly one of 29 sub-assets inside the eight asset domains, percentile-ranked against the 99-county peer set, and separately tested against the Georgia statewide value.

Four findings

1. **The reported constraints are real, and the score is credible.** Median asset score is 50.8 on a 0–100 peer scale. Three counties score as overperforming, 31 as solid, 47 at risk and 18 as underperforming. The ranking places Chatham, Thomas, Glynn, Lowndes and Tift at the top and Baker, Taliaferro, Twiggs, Wilcox and Wheeler at the bottom — an ordering that matches what practitioners in the region would independently produce, which is the first evidence that the instrument measures what it claims to.
2. **Scale is itself an asset — and that has to be separated out.** The raw composite correlates 0.80 with log population. That is partly a real finding (agglomeration is genuine in rural economies) and partly an artefact. We therefore publish a second, size-adjusted score — the residual of each indicator on log population — that correlates 0.07 with population. The two answer different questions and rank counties very differently. Richmond, Muscogee, Columbia, Houston and Bibb all fall more than 55 places when size is controlled for; Clay, Webster, Glascock, Taliaferro and Quitman all rise more than 45.
3. **Constraints bundle, decisively.** Against a null model that holds each constraint's prevalence fixed but scatters it independently across counties, the observed spread of constraint counts is 2.4 times too large ($p < 0.0001$). 71 of 406 constraint pairs co-occur significantly, against 20 expected by chance. Five distinct bundles emerge — though they are tendencies on a gradient, not crisp types.
4. **There is no universal sequence, but there is a threshold.** Cumulative-scale analysis returns a Loevinger's H of 0.057 (95% CI 0.036–0.085) — far below the 0.30 needed for a cumulative order. Counties do not acquire constraints in a fixed order, so no single "fix this first" rule holds region-wide. But outcomes break sharply at about ten constrained sub-assets, and the marginal value of relieving one constraint falls by roughly four-fifths above that break. Sequencing works below the threshold; it does not above it.

What this means for Bridges' systems-integration hypothesis

The hypothesis holds, with a condition attached rather than universally. In counties carrying fewer than about ten constraints, constraints behave more independently and a targeted single-asset intervention is defensible and efficient. In counties above the threshold — 83 of the 99 counties, on the absolute constraint count —

constraints are mutually reinforcing and relieving one moves outcomes very little. For those counties a coordinated package is the evidence-based approach, and a sequential one is not. The practical implication is that the delivery model should be triaged by constraint load rather than applied uniformly.

1. Study design

1.1 Geography

The study area is every county in Georgia Regional Commissions 6 through 12: Middle Georgia (11 counties), Central Savannah River Area (13), River Valley (16), Heart of Georgia–Altamaha (17), Southwest Georgia (14), Southern Georgia (18) and Coastal (10) — 99 counties in total. Membership was verified against the Georgia Department of Community Affairs regional boundary definition; the twelve commissions sum to Georgia's 159 counties as a cross-check.

The full seven-commission set was used rather than a narrower five-commission cut because the bundling and clustering analysis in Sections 4 and 5 needs the sample size. With 99 counties and 29 constraint dimensions the analysis is already at the edge of what the data will support; a smaller frame would not sustain it.

1.2 The asset framework

The eight domains and their component assets were specified by the client. Each was decomposed into measurable sub-assets, and 92 indicators were assigned to them — with each indicator belonging to exactly one sub-asset. That one-to-one assignment matters: if an indicator contributed to two sub-assets, those sub-assets would correlate by construction and the bundling analysis would find structure that is an artefact of the framework rather than a fact about the counties.

Domain	Sub-assets measured	Indicators
Workforce	Labour force participation · Employment · K-12 system capacity · Employer engagement · Workforce retention	16
Housing	Affordability · Supply · Vacancy · Permitting · Workforce commuting	17
Infrastructure	Water/sewer capacity · Broadband access · Transportation assets · Industrial site readiness	12
Healthcare	Provider availability · Hospital & health sector employment · Access indicators · Health outcomes	17
Local capacity	Staffing · Planning capacity	5
Demographics	Population change · Migration · Aging · Educational attainment	10
Industry & innovation	Business formation · Cluster concentration · Entrepreneurship · Knowledge & research services	10
Civic & regional	Civic participation	3

29 sub-assets scored; 8 further sub-assets from the original framework could not be measured — see Section 7.

1.3 Scoring method

Every indicator is oriented so that higher always means better, then percentile-ranked within the 99-county peer set. Sub-asset scores are the mean of their indicator percentiles; domain scores are the mean of their sub-asset scores; the composite Asset Quality Score is the unweighted mean of the eight domain scores.

Equal domain weighting is a choice, not a finding. It is transparent and avoids embedding a theory of which domain matters most, but it does mean a domain measured by one sub-asset (Civic & Regional) carries the same weight as one measured by five (Workforce). Users who want a different weighting can recompute from the published sub-asset scores. Domain coverage is reported alongside every domain score in the dashboard so this is visible rather than hidden.

1.4 Two scores, not one

The unadjusted composite correlates 0.80 with log population. Some of that is real — larger rural counties genuinely have more providers, more employers, more sectors and better broadband — and some is mechanical. Presenting only the raw score would hand the client a population ranking dressed as an asset assessment.

We therefore compute a second score. For each indicator, the peer percentile is regressed on log population and the residual re-ranked. The resulting size-adjusted composite correlates 0.07 with population, and its mean is flat across population quintiles (49.6, 48.9, 49.7, 52.5, 51.7 from smallest to largest), confirming that a linear adjustment is adequate — a quadratic term explains only 2.2% more.

Question you are asking	Score to use
How much asset base does this county actually have? Where should a fixed-size investment go furthest in absolute terms?	Absolute score
Given this county's scale, is it over- or under-performing? Which counties punch above or below their weight?	Size-adjusted score
Which constraints co-occur, and do they bundle?	Size-adjusted — otherwise every constraint correlates with every other simply because all correlate with population

1.5 The absolute test, and the four constraint states

Peer ranking alone cannot distinguish a healthy county from the best of a struggling group. Each raw value is therefore also compared with the Georgia statewide figure, expressed in peer standard deviations. Combining the two tests gives four states.

State	Definition	What it tells you
Confirmed constraint	Bottom third of peers AND more than 0.5 SD below Georgia	Both tests agree. The clearest case for action.
Relative gap	Bottom third of peers, but not materially below Georgia	Behind the neighbours, not behind the state. Often a local, tractable problem.
Region-wide constraint	Not in the bottom third locally, but well below the state figure	The whole peer group is behind Georgia. Peer ranking hides this; it usually calls for a regional or

State	Definition	What it tells you
		state-level response, not a county one.
Not constrained	Neither test triggers	—

The region-wide category is the analytically important one. It is why the client chose a dual benchmark, and it changes the reading of several domains. Knowledge and research services is constrained in 98% of counties but confirmed in only 27% — the other 71% are region-wide. The same pattern holds for health outcomes (88% constrained, 67% region-wide), labour force participation (79%, 60%) and educational attainment (88%, 59%). These are not county failures. They are structural features of the study area that no individual county can fix alone.

2. Asset quality across the study area

Median asset score is 50.8 with a standard deviation of 10.4; the range runs from 29.2 (Baker) to 75.4 (Chatham). On the size-adjusted score the spread narrows (SD 7.6), as expected once the population gradient is removed.

Band	Absolute score	Size-adjusted
Overperforming (70+)	3	0
Solid (55–70)	31	30
At risk (40–55)	47	61
Underperforming (25–40)	18	8
Severely constrained (<25)	0	0

Counties by performance band, absolute and size-adjusted scores. Bands are cut on a peer-relative scale, so no county falls below 25 on either measure.

2.1 Strongest and weakest counties

Strongest (absolute)	Score	Weakest (absolute)	Score
Chatham (Coastal)	75.4	Baker (Southwest GA)	29.2
Thomas (Southwest GA)	72.0	Taliaferro (CSRA)	30.3
Glynn (Coastal)	71.2	Twiggs (Middle GA)	30.8
Lowndes (Southern GA)	66.6	Wilcox (HOGA)	31.0
Tift (Southern GA)	65.3	Wheeler (HOGA)	31.7
Dougherty (Southwest GA)	64.8	Hancock (CSRA)	32.7
Bulloch (Coastal)	64.5	Stewart (River Valley)	34.5
Monroe (Middle GA)	64.4	Quitman (River Valley)	35.3

This ordering is a validity check as much as a result. The top of the list is the region's trade and service centres — Savannah, Thomasville, Brunswick, Valdosta, Tifton, Albany, Statesboro — plus one fast-growing Macon suburb. The bottom is the persistently-poor Black Belt and south-central counties. Neither list would surprise anyone working in the region, which is the point: an instrument that produced a different ordering would be suspect.

2.2 Counties that punch above and below their weight

The gap between the two scores is where the actionable insight sits. Comparing each county's rank on the absolute and size-adjusted measures:

Punching above their weight	Pop.	Abs.	Adj.	Punching below	Pop.	Abs.	Adj.
Clay	2,815	44.7	59.1	Richmond	206,303	63.3	42.5
Webster	2,315	38.8	55.1	Muscogee	201,830	63.6	43.7
Glascock	3,008	43.5	56.5	Columbia	167,472	62.8	43.8
Taliaferro	1,620	30.3	50.9	Houston	174,897	61.9	43.7
Quitman	2,323	35.3	51.5	Bibb	157,056	63.8	46.7
Schley	4,503	47.3	57.6	Liberty	68,607	58.2	47.5
Clinch	6,871	49.3	57.2	Effingham	73,148	58.5	48.4
Randolph	6,085	50.2	59.0	Chatham	307,336	75.4	52.7

The pattern on the right is worth the client's attention. Every urban county in the study area under-delivers relative to its scale. Richmond, Muscogee, Columbia, Houston and Bibb all fall more than 55 rank positions once population is controlled for. This is not a regression artefact — mean size-adjusted score is flat across population quintiles, so large counties are not systematically penalised. It is a finding: south Georgia's cities are not converting their scale into the asset quality that scale usually delivers, and they are doing worse at it than the region's small counties are doing at their own scale.

2.3 By Regional Commission

Commission	Work-force	Housing	Infra.	Health	Local cap.	Demog.	Industry	Civic	Absolute	Adjusted
Coastal	58.9	56.7	61.0	62.9	49.2	77.4	60.1	52.2	59.8	52.0
Middle Georgia	46.8	51.5	56.6	59.2	46.4	57.6	53.3	45.6	52.1	48.8
Southwest Georgia	52.1	50.1	45.6	49.8	55.8	42.0	54.9	56.5	50.9	51.8
CSRA	48.9	49.5	46.4	50.8	55.6	46.8	51.0	53.8	50.4	52.0
Southern Georgia	56.7	52.7	50.5	46.1	50.5	52.3	48.7	42.7	50.0	50.5
Heart of GA–Altamaha	50.6	44.6	49.4	45.3	44.3	46.6	49.3	47.9	47.3	48.3
River Valley	40.2	46.0	47.6	45.1	51.6	40.8	41.7	56.4	46.2	50.6

On the absolute score the commissions separate clearly: Coastal leads by nearly 14 points, River Valley and Heart of Georgia–Altamaha trail. On the size-adjusted score they converge to within four points of each other. Almost all of the between-commission difference in asset quality is a difference in the size of their counties, not in how well those counties perform for their size. That is a materially different message for a regional director than the raw ranking gives.

3. Which assets are constrained

On the absolute definition, counties carry a median of 17 constrained sub-assets out of 29 (mean 15.9, range 3 to 25). On the size-adjusted definition the median is 7 (mean 7.3, range 1 to 17). Both numbers matter: the first says how much of the asset base is weak in absolute terms, the second how much is weak relative to what the county's scale should deliver.

3.1 Constraint prevalence across the study area

Sub-asset	Any	Confirmed	Region-wide	Size-adj.
Knowledge & research services	98%	27%	71%	33%
Health outcomes	88%	21%	67%	21%
Educational attainment	88%	29%	59%	29%
Broadband access	81%	32%	48%	30%
Employer engagement	81%	37%	43%	30%
Labor force participation	79%	19%	60%	22%
Business formation	78%	20%	58%	18%
Provider availability	74%	18%	56%	26%
Water/sewer capacity	74%	23%	51%	27%
Employment	68%	21%	46%	24%
Population change	65%	32%	32%	32%
Vacancy	64%	29%	34%	29%
Entrepreneurship	62%	23%	38%	24%
Transportation assets	62%	26%	35%	27%
Permitting	61%	28%	32%	30%
Aging	60%	30%	29%	25%
Access indicators	58%	28%	29%	27%
Workforce retention	44%	23%	21%	16%
Industrial site readiness	42%	26%	16%	25%
Hospital & health sector employment	41%	36%	5%	23%
Supply	33%	11%	22%	11%
Migration	32%	25%	7%	27%
Cluster concentration	31%	15%	16%	24%

Sub-asset	Any	Confirmed	Region-wide	Size-adj.
Workforce commuting	28%	19%	0%	24%
K-12 system capacity	27%	12%	1%	28%
Affordability	23%	2%	0%	18%
Staffing	20%	4%	0%	25%
Civic participation	16%	12%	0%	20%
Planning capacity	15%	0%	0%	23%

Share of the 99 counties constrained in each sub-asset. "Any" is the absolute definition (confirmed + region-wide + relative gap); "Size-adj." is the bottom third after removing the population gradient.

3.2 Reading the table

- **Near-universal, but mostly structural.** Knowledge and research services (98%), health outcomes (88%), educational attainment (88%), broadband (81%) and labour force participation (79%) are constrained almost everywhere — but in each case the majority of those flags are region-wide rather than confirmed. These are the gaps between rural south Georgia and the state, not between one county and its neighbours.
- **Where county-level action bites.** Employer engagement (37% confirmed), hospital and health-sector employment (36%), population change (32%), broadband (32%) and aging (30%) have the highest confirmed rates — the county is behind both its peers and the state. These are the sub-assets where a county-level intervention has the clearest case.
- **The apparent bright spots deserve scepticism.** Housing affordability is constrained in only 23% of counties and local government staffing in 20%. Affordability looks good because incomes and house prices are both low — the median home-value-to-income ratio in the study area is 2.5 against 3.7 statewide — not because housing is abundant or good. Staffing looks adequate on a per-capita basis because small counties carry fixed administrative overhead spread across few residents; on absolute headcount the same counties have almost no capacity. Both measures are reported in the dashboard with their absolute counterparts so the reader can see the difference.
- **Planning capacity is the largest gap between the two definitions.** 15% constrained on the absolute measure, 23% on the size-adjusted one — the only sub-asset where the size-adjusted rate materially exceeds the absolute rate for that reason. Small counties look well-staffed per head and are not.

4. Do constraints bundle?

Yes — and the result is not marginal. This is the central empirical claim of the memo, so the test is worth stating precisely.

4.1 The test

The null hypothesis is that constraints occur independently: some counties happen to have a broadband problem, others a housing problem, and there is no tendency for the same county to have both. To test it we build a null distribution in which each sub-asset's constrained counties are re-drawn at random across the 99 counties, holding that sub-asset's prevalence exactly fixed. This breaks any tendency to co-locate while preserving how common each constraint is. 20,000 draws.

Statistic	Observed	Null (mean)	Null 95% interval	p
Variance in constraints per county	12.95	5.42	4.05 – 7.01	< 0.0001
Share of counties with 15+ of 29 constraints	5.1%	0.2%	0.0% – 1.0%	< 0.0001
Share of counties with 3 or fewer constraints	15.2%	4.5%	1.0% – 8.1%	< 0.0001
Constraint pairs co-occurring at p<0.05	71 of 406	20 expected	—	< 1e-15

Constraints carry 2.4 times more variance across counties than chance allows. Counties with fifteen or more constraints occur at twenty-five times the null rate, and counties with almost none occur at three times the null rate. Both tails are fat — constraints concentrate.

4.2 The strongest co-occurrences

Constraint A	Constraint B	φ
Educational attainment	Access indicators	0.55
Migration	Population change	0.55
Hospital & health sector employment	Employer engagement	0.47
Workforce commuting	Employer engagement	0.45
Employer engagement	Workforce retention	0.43
Population change	Employment	0.42
Hospital & health sector employment	Workforce retention	0.41
Provider availability	Employer engagement	0.41
Permitting	Vacancy	0.40
Educational attainment	Knowledge & research services	0.39

Phi coefficients between binary constraint flags, size-adjusted definition. All pairs significant at p<0.05 against the permutation null.

These pairings are readable, which is itself reassuring. Educational attainment travels with healthcare access; migration with population change; hospital and health-sector employment with employer engagement and with workforce retention; permitting with vacancy. The employer-engagement node is involved in four of the top ten — the local employer base is where the workforce, healthcare and commuting constraints meet.

4.3 Five bundles — but they are tendencies, not types

Ward clustering on the 99 × 29 constraint matrix produces five interpretable groups. Before describing them, the honest caveat: silhouette scores run 0.10–0.18 and bootstrap agreement 0.24–0.36. Counties sit on a gradient with soft syndromes, not in crisp categories. Solutions from four to six clusters give substantively the same groups. Treat the bundles as a way of organising the analysis, not as a classification any individual county firmly belongs to.

Bundle	n	Median pop.	Score	Constraints	Poverty	Pop. change
Deep multi-system distress	7	12,349	37.4	13.7	20.2%	−1.0%
Growth outpacing capacity	20	19,617	49.2	8.0	17.5%	+4.5%
Demographic contraction	19	15,019	50.8	7.8	24.0%	−2.3%
Exported workforce	14	11,493	47.9	7.9	20.4%	+2.2%
Low constraint load	39	19,834	53.9	5.2	19.4%	+1.2%

Constraints column is the mean size-adjusted count of 29; study-area average is 7.3.

Deep multi-system distress (7 counties)

Every system strained at once. These seven carry roughly twice the study-area constraint load, and the over-represented constraints span workforce, local government, healthcare and housing simultaneously. This is the bundle the systems-integration argument was written for — and the one where, on the evidence in Section 5, a sequential approach is least likely to work.

Growth outpacing capacity (20 counties)

The fastest-growing bundle at +4.5% population, and the constraints sit in service and civic capacity rather than in jobs or housing demand. Provider availability, planning capacity, entrepreneurship and civic participation are all over-represented; population change and permitting are under-represented. The group mixes metro-edge counties (Houston, Columbia, Effingham, Liberty) with small counties adding population from a very low base. The policy read is capacity catching up with demand, not demand needing stimulation.

Demographic contraction (19 counties)

Losing population and migrants, with permitting and vacancy problems following the decline and weak health outcomes alongside. Poverty averages 24%, the highest of the five bundles. Notably this group contains two of the region's larger cities (Macon–Bibb and Albany–Dougherty) alongside small rural counties, which is a reminder that contraction is not only a small-county phenomenon.

Exported workforce (14 counties)

Residents commute out to work. Workforce commuting is over-represented by 54 points, and it travels with a weak local employer base, low hospital and health-sector employment and an older age structure. Housing and population are comparatively stable — these are counties whose economies are effectively hosted somewhere else. The intervention question here is whether to build a local employer base or to improve the terms on which residents access the external one.

Low constraint load (39 counties)

The lowest constraint load once size is accounted for. No constraint is over-represented by more than 15 points. This is a performance group rather than a settlement type: it contains the large regional trade centres and also small counties that outperform what their scale predicts (Clay, Webster, Glascock, Quitman, Taliaferro). Remaining gaps are specific rather than systemic, and targeted single-asset work is appropriate here.

5. Sequencing or simultaneity?

This is the question that decides the delivery model, and it has three parts: is there a universal order, is there a threshold, and are there local prerequisites? The answers are no, yes, and yes.

5.1 There is no universal sequence

If constraints accumulated in a consistent order — if, say, a county always lost its hospital before it lost its employer base — the constraint set would form a cumulative scale. Mokken scale analysis measures this directly through Loevinger's coefficient H, which counts how often the observed pattern violates a consistent ordering.

H = 0.057 (bootstrap 95% CI 0.036–0.085). The conventional thresholds are 0.30 for a weak scale, 0.40 for medium and 0.50 for strong. This is an order of magnitude below the weakest of them. Counties do not acquire constraints in a fixed order.

The practical consequence is direct: there is no defensible region-wide "tackle this first" prescription. A programme built on one — workforce first, or housing first, or broadband first — would be applying an order the data does not support to counties whose constraint profiles differ substantially.

5.2 But outcomes break at about ten constraints

Constraint count does predict outcomes, and not linearly. Fitting a segmented regression across candidate breakpoints:

Outcome	Breakpoint	80% bootstrap CI	Mean below	Mean above	Segmented vs linear
Median household income	10	9 – 17	$70,433	$50,016	F = 5.16, p = 0.007
Poverty rate	10	9 – 21	14.6%	21.1%	F = 1.62, p = 0.20
Population change 2020–24	9	9 – 20	+6.3%	+0.4%	F = 2.37, p = 0.10

All three outcomes locate their break in the same narrow region — nine or ten constraints out of twenty-nine — which is more persuasive than any one of them alone. The income break is statistically clean (p = 0.007); poverty and population change point the same way with wider confidence intervals. The honest statement is that the threshold is real and its location is estimated at ten with meaningful uncertainty, not that ten is a precise cut point.

The magnitude is large. Counties at or below the break have median household income of $70,433; above it, $50,016 — a gap of $20,417, or 29%.

5.3 And the return to relief collapses above the threshold

This is the finding that speaks most directly to Bridges' hypothesis. Estimating the marginal association between constraint count and median income separately within constraint bands:

Counties carrying	n	Marginal value of one fewer constraint
9 to 14 constraints	26	+$2,611 in median household income
15 or more constraints	64	+$565 in median household income

Relieving a single constraint inside a deeply constrained county is worth roughly one-fifth of what it is worth in a moderately constrained one. That is the strongest evidence in this study for addressing bundles together: where constraints are dense, they are mutually reinforcing, and single-asset interventions have little traction. Where they are sparse, they behave more independently and targeted work pays.

Caveat. This is a cross-sectional association across counties, not a causal estimate of what happens when a given county relieves a constraint. It is consistent with mutual reinforcement; it does not prove it. Section 8 sets out the panel design that would.

5.4 Local prerequisites do exist

Although there is no global order, specific pairs show strong asymmetry — B rarely occurs without A, while A frequently occurs without B. Aggregating that asymmetry across all pairs gives a net upstream score for each constraint.

Most upstream (others depend on them)	Net	Most downstream (depend on others)	Net
Knowledge & research services	+2.71	Workforce retention	-2.86
Employer engagement	+2.27	Business formation	-2.74
Population change	+2.02	Supply	-2.45
Permitting	+1.51	Civic participation	-1.35
Vacancy	+1.37	Affordability	-1.34
Educational attainment	+1.34	Labor force participation	-1.18

Read this as: a county with a housing-supply constraint almost always also has a healthcare-access, permitting and vacancy constraint, but not the reverse. Housing supply is therefore a late, conditional symptom rather than a root cause — and treating it in isolation is unlikely to work. Conversely, knowledge and research services, employer engagement, population change and permitting appear early and broadly; other constraints tend to sit downstream of them.

Network centrality on the co-occurrence graph tells a compatible story. The most central nodes are Employer engagement, Staffing, Workforce retention, Hospital & health sector employment — employer engagement is the single most embedded constraint in the structure, which is consistent with its appearance in four of the ten strongest pairwise co-occurrences.

5.5 The answer, stated plainly

Bridges' systems-integration hypothesis holds for the counties it was formulated about, and matters less for the rest. Three results together:

- No universal sequence exists (H = 0.06), so a fixed region-wide ordering of interventions is not defensible.
- Outcomes break at roughly ten constraints, so the relevant distinction is not which constraint but how many.
- Above that break the marginal return to relieving one constraint falls by about 80%, so in dense-constraint counties a package is required and a sequence will underdeliver.

The operational implication is a triaged delivery model rather than a uniform one:

County profile	Approach	Counties in study area
Below the threshold — 10 or fewer constrained sub-assets of 29, absolute definition	Targeted single-asset work. Use the upstream ranking in 5.4 to choose which. Efficient and defensible here.	16 counties (16%)
Above the threshold	Coordinated package across the county's specific bundle. Single-asset work will not move outcomes enough to justify itself.	83 counties (84%) — including the whole of the "deep multi-system distress" bundle
Any county	Region-wide constraints (Section 1.5) need a regional or state response regardless of county profile — no county can close them alone.	All 99

Note the asymmetry in those counts. Only sixteen of the ninety-nine counties sit below the threshold on the absolute measure. If the finding in 5.3 holds up under the panel test proposed in Section 8, it implies that single-asset intervention is the efficient design for a small minority of the study area and the integrated package is the right default for the rest — which is close to the inverse of how rural development assistance is typically delivered.

6. Limitations

- **Cross-sectional.** Every result here describes structure at a point in time. The sequencing analysis identifies which constraints are structurally upstream, not which came first in any county's history. Causal ordering requires panel data.
- **Soft clusters.** The five bundles have silhouette 0.10–0.18 and bootstrap agreement 0.24–0.36. They are a useful organising device, not a classification. A county near a boundary could reasonably be assigned to either side.
- **The threshold is estimated, not measured.** Ten constraints is a point estimate with a bootstrap interval of 9–17 on the cleanest outcome. The existence of a break is well supported; its exact location is not.
- **Equal domain weights.** A defensible default, but a choice. Domains measured by one sub-asset carry the same weight as those measured by five.
- **Eight sub-assets are unmeasured.** Section 7. Two of them — grant awards and audit compliance — are among the most policy-relevant in the whole framework.
- **Broadband is measured as adoption, not availability.** We use ACS household subscription rather than FCC availability. For economic development purposes adoption is arguably the better measure, but they are not the same thing and a county with good availability and poor adoption will score as constrained here.
- **Some proxies are weak and are labelled as such.** Every indicator carries an evidence grade in the dashboard: direct measure, strong proxy, or weak proxy. Employer engagement and industrial site readiness rest on weak proxies and should be read with that in mind.

7. What could not be measured

Eight sub-assets from the original framework have no defensible public county-level measure. They are excluded from the score rather than filled with a weak proxy, because a bad proxy in a composite index is worse than an acknowledged gap.

Domain / sub-asset	Why it could not be measured	Recommended source
Local capacity — Grant awards	The Federal Audit Clearinghouse holds exactly this data but requires API credentials; USAspending's county aggregation is not available through a public GET endpoint.	FAC API key, or the state's own grant administration records
Local capacity — Audit compliance	The Georgia Department of Audits and Accounts publishes a Local Government Compliance Dashboard with precisely the needed data, but it renders inside a Power BI frame with no machine-readable export.	Direct request to DOAA, or browser-based extraction
Workforce — Training capacity	No public county-level data on TCSG enrollment, completions or campus capacity. County Business Patterns captures private trade schools only, and 92% of counties report zero.	TCSG enrollment and completion files by campus
Workforce — Postsecondary presence	The Census of Governments attributes no higher-education employment to counties, because USG and TCSG institutions are state-operated and recorded at state level.	USG and TCSG campus roster with enrollment
Infrastructure — Power availability	No public county-level measure of generation, transmission or interconnection capacity exists. Municipal electric employment covers only 42% of counties and reflects ownership structure, not availability.	Georgia EMC and Georgia Power service data; survey
Civic & regional — Collaboration	The only available proxy — local government units per capita — is mechanically the inverse of population (r = 0.88 with log population). It measures size, not collaboration.	Field survey
Civic & regional — Shared services	No public register of intergovernmental service agreements or joint authorities.	Field survey; RC records
Civic & regional — Regional organisations	Constant within each Regional Commission, so there is no county-level variation to score.	n/a — structural, not variable

Two of these are worth pursuing immediately. Audit compliance and grant awards are the strongest available signals of local government capacity, they are both named in the client's original framework, and both exist in a usable form behind an access barrier rather than not existing at all. Adding them would materially strengthen the Local Capacity domain, which is currently the thinnest in the index.

8. Recommended next steps

5. **Close the two capacity gaps.** Obtain the DOAA compliance data and a Federal Audit Clearinghouse extract. This is a matter of access, not of data existence, and it would take the Local Capacity domain from two sub-assets to four.
6. **Field-verify the bundles in three counties per bundle.** The clusters are soft. Structured interviews in fifteen counties — three from each bundle, chosen at the centre rather than the boundary — would

establish whether the syndromes describe what practitioners actually see, and would either validate the typology or replace it with a better one.

7. **Build the survey instrument for the unmeasured sub-assets.** Employer engagement, shared services, collaboration, training capacity and planning capacity all need primary collection. A single instrument administered through the Regional Commissions would cover all five and could be repeated annually.
8. **Construct the panel.** The threshold and marginal-return findings are the ones that would change practice, and both are currently cross-sectional. Nearly every indicator here is available back to 2010 or earlier. A ten-year county panel would allow a fixed-effects test of whether relieving a constraint actually improves outcomes, and whether that effect is conditional on the county's remaining constraint load — which is the direct causal test of the systems-integration hypothesis.
9. **Stratify the delivery model by constraint load.** Rather than offering the same programme design to all 99 counties, use the constraint count as a triage variable: targeted assistance below the threshold, integrated packages above it, and regional or state-level advocacy for the region-wide constraints no county can close alone.

Appendix A. Data sources

Source	Vintage	What it contributes
American Community Survey, 5-year	2019–2023	40 detailed tables: labour force, educational attainment, housing cost burden, vacancy, tenure, stock age, internet subscription, commuting, insurance coverage, disability, poverty, age structure, migration, self-employment
County Business Patterns	2023 and 2013	Employment, establishments and establishment size structure by sector; location quotients; ten-year establishment change
Census of Governments, Employment	2022	Full census of 1,258 Georgia government units: K-12 staffing, water and sewer systems, community development, public health, hospitals, financial administration
LEHD LODES 8	2023	County-to-county commuting flows: workforce retention, out-commuting, jobs per resident worker
BLS Local Area Unemployment Statistics	2019–2025	Labour force, employment and unemployment, annual averages
Census Population Estimates	2020–2024	Population change, net and domestic migration rates, natural change, births and deaths
Census Building Permits Survey	2015–2024	Residential permitting by county and structure type
CDC PLACES	2024 release	Age-adjusted prevalence: diabetes, obesity, hypertension, poor mental health, depression, smoking, uninsured, routine checkup
CMS Hospital General Information	Current	All 149 Georgia hospitals with type, ownership and emergency services
TIGER primary roads and Gazetteer	2024	Distance from county internal point to nearest Interstate

Appendix B. A note on method choices

Why percentile ranks rather than z-scores

Several indicators are strongly skewed — establishment counts, hospital presence, distance to interstate. Percentile ranking is robust to that skew and makes the composite interpretable as "position among peers" without imposing a distributional assumption. The cost is loss of information about magnitude, which is why the absolute test against Georgia is reported alongside every score.

Why the size adjustment is linear in log population

Mean size-adjusted score is flat across population quintiles (49.6 to 52.5), and adding a quadratic term reduces mean residual variance by only 2.2%. The linear form is adequate and simpler to explain.

Why the permutation null shuffles columns independently

An earlier version used a curveball algorithm preserving both row and column marginals. That is the wrong null here: it preserves the row-sum distribution exactly, so it cannot test whether row sums are more dispersed than chance. Independent column shuffling preserves each constraint's prevalence — which is what we want to hold fixed — while breaking the tendency to co-locate, which is what we want to test.

Why hospital employment is not used directly

County Business Patterns excludes government entities, and 48 of Georgia's 149 hospitals are county hospital authorities. The Census of Governments captures authority employment, but many authorities lease operations to non-profit operators and report zero staff. Combining both sources still leaves 29 counties with a hospital but no measurable hospital employment. We therefore score facility presence from CMS records — which is complete and authoritative — and use total healthcare-sector employment, which has only 5% zero coverage, for the employment dimension.